

Ranking the disease burden of 14 pathogens in food sources in the United States using attribution data from outbreak investigations and expert elic...

Understanding the relative public health impact of major microbiological hazards across the food supply is critical for a risk-based national food safety system. This study was conducted to estimate the U.S. health burden of 14 major pathogens in 12 broad categories of food and to then rank the resulting 168 pathogen-food combinations. These pathogens examined were *Campylobacter*, *Clostridium perfringens*, *Escherichia coli* O157:H7, *Listeria monocytogenes*, norovirus, *Salmonella enterica*, *Toxoplasma gondii*, and all other FoodNet pathogens. The health burden associated with each pathogen was measured using new estimates of the cost of illness and loss of quality-adjusted life years (QALYs) from acute and chronic illness and mortality. A new method for attributing illness to foods was developed that relies on both outbreak data and expert elicitation. This method assumes that empirical data are generally preferable to expert judgment; thus, outbreak data were used for attribution except where evidence suggests that these data are considered not representative of food attribution. Based on evaluation of outbreak data, expert elicitation, and published scientific literature, outbreak-based attribution estimates for *Campylobacter*, *Toxoplasma*, *Cryptosporidium*, and *Yersinia* were determined not representative; therefore, expert-based attribution were included for these four pathogens. Sensitivity analyses were conducted to assess the effect of attribution data assumptions on rankings. Disease burden was concentrated among a relatively small number of pathogen-food combinations. The top 10 pairs were responsible for losses of over \$8 billion and 36,000 QALYs, or more than 50 % of the total across all pairs. Across all 14 pathogens, poultry, pork, produce, and complex foods were responsible for nearly 60 % of the total cost of illness and loss of QALYs.